

Dynamics of the EUR/USD Exchange Rate during 2022

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Abstract

This paper examines the 2022 dynamics of the EUR/USD exchange rate, contributing insights to the understanding of global exchange-rate variations. Exploring factors such as interest parity, politics, central bank policies, and impactful announcements, we aim to provide a comprehensive view of the forces influencing this currency pair during this period.

1 Introduction

The foreign exchange market lies at the heart of the world economy. Since the end of the Bretton Woods system in the early 1970s, variations in exchange rates between the world's major currencies have been substantial, affecting household, corporate, and government decision making across the world. Understanding the source and consequences of exchange-rate variations has thus remained a prominent goal in international economics through the years. Over the last decade, the role of exchange rate intervention has taken on greater prominence in policy discussions. Key questions have centered on the effectiveness of exchange rate intervention as a policy instrument in a central bank's arsenal—that is for smoothing macroeconomic volatility, applying political pressure, and helping attain inflation targets at times when the effectiveness of conventional monetary policy instruments reaches limits.

This paper aims to contribute fresh insights by analysing the dynamics of the EUR/USD exchange rate throughout the year 2022. Our goal is to enhance the understanding of the dominant factors influencing exchange rates in today's global economic context.

To achieve this objective, we will dive into key aspects such as interest parity, the influence of central bank policies, and the impact of pivotal announcements or news events. By exploring these dimensions, we try to offer a comprehensive perspective on the forces driving the dynamics of the EUR/USD exchange rate during the specified timeframe.

2 Background

The emergence of the COVID-19 pandemic in 2020 prompted central banks, such as the Federal Reserve (Fed) and the European Central Bank (ECB), to enact immediate and comprehensive monetary policy measures. These actions were aimed at stabilizing financial markets, ensuring liquidity, and mitigating the economic fallout caused by the unprecedented global health crisis. The following overview examines the specific measures implemented by the Fed and the ECB from 2020 to 2021 in response to the challenges posed by the COVID-19 pandemic.

2.1 Federal Reserve (Fed) - United States (2020-2021):

The Fed implemented a multifaceted response to the COVID-19 crisis. Monetary policy measures included a 150 basis points cut in the federal funds rate to near zero, coupled with forward guidance on keeping rates low until specific economic conditions were met. Quantitative Easing (QE) involved substantial purchases of government and mortgage-backed securities, which increased the money supply shown in Fig. 6 and 9. The Fed also established pandemic-era facilities, such as the Primary Dealer Credit Facility and the Money Market Mutual Fund Liquidity Facility, to stabilize financial markets. Direct lending to banks and major corporate employers, as well as programs like the Main Street Lending Program and Commercial Paper Funding Facility, aimed to ensure the flow of credit to businesses. Support was extended to households through the Term Asset-Backed Securities Loan Facility. Additionally, the Fed addressed state and municipal borrowing through the Municipal Liquidity Facility.

2.2 European Central Bank (ECB) (2020-2021):

The ECB responded to the COVID-19 crisis with targeted monetary policy measures. The Pandemic Emergency Purchase Programme (PEPP), announced in March 2020, played a crucial role in stabilizing markets and addressing fragmentation. The recalibration of the Targeted Longer-Term Refinancing Operations (TLTRO III) in April 2020 provided favorable interest rates to support credit provision. Additional measures included Longer-Term Refinancing Operations (LTROs) and Pandemic Emergency LTROs (PELTROs) for immediate liquidity support. The ECB's actions aimed to ease financial conditions, support credit flow to the real economy, and contribute to medium-term growth and inflation. The PEPP, with its flexibility, played a pivotal role in restoring the transmission of monetary policy across the euro area as we can see in Fig. 7 and 8. The overall impact of these measures included a significant reduction in sovereign yields and support for the euro area's economic outlook.

2.3 Inflation Surge (2021-2022):

In response to the COVID-19 crisis, the Federal Reserve and the European Central Bank swiftly implemented comprehensive measures to stabilize economies. From interest rate adjustments to targeted lending programs, these actions successfully averted financial crises and supported credit flow. As the recovery progressed, a noteworthy outcome surfaced—a global surge in inflation beginning in mid-2021.

In 2021, inflation rates reached historic highs in the US and in the eurozone, as shown in Fig. 4 and Fig. 5. This phenomenon was attributed to a multitude of factors, including pandemic-induced economic disruptions, supply chain challenges, and extensive fiscal and monetary stimuli enacted in 2020 and 2021.

3 Graphical Analysis

3.1 Graph Interpretation

The graph in Fig. 1 presents the exchange rate trend of the Euro against the US Dollar for the year 2022. It shows a dynamic yet generally declining trajectory for the Euro. The year begins with the Euro at a position of relative strength against the Dollar, indicated by the exchange rate being above 1.0. As the year progresses, the Euro experiences a downward trend against the Dollar, with the exchange rate descending beneath the 1.0 mark, suggesting a weakening of the Euro or a strengthening of the Dollar.

This decline is punctuated by periods of volatility, where the exchange rate fluctuates but continues on a downward path until it reaches its nadir around October at 0.96. After this low point, the Euro began to recover, showing an upward trend in the latter part of the year to finally reach 1.05 at the end of the year.

3.2 Theoretical Analysis

The theoretical analysis behind the EUR/USD exchange rate fluctuations can be discerned through mathematical equations that link interest rates to the equilibrium in the money market.

The Interest Rate Parity (IRP) and Money Market conditions are fundamental to this understanding. The IRP condition posits that the expected exchange rate movement between two economies is influenced by their interest rate differential, encapsulated by the equation $E_{s/\$}^e = E_{s/\$} \times \frac{(1+i_e)}{(1+i_s)}$, where $E_{s/\e denotes the expected future spot exchange rate, $E_{s/\$}$ is the current spot rate, i_e the Eurozone interest rate, and i_s the US interest rate. According to this model, a higher interest rate in the Eurozone compared to the US suggests an anticipated depreciation of the Euro against the Dollar. The Money Market condition further states that $M/P = L(i, Y)$, linking the money supply (M), price level (P), interest rate (i), and real output (Y). We can then see that money supply affects the interest rate which influences the exchange rate.

These theoretical frameworks provide a comprehensive view of the dynamic interplay between monetary policy changes and EUR/USD exchange rate trends observed in 2022. The detailed implications of these equations will be further explored in the subsequent sections of this analysis.

3.3 Global Context Influence

As already said before, the fall in the EUR/USD exchange rate in 2022 can be explained by several key factors linked to divergent monetary policies, economic conditions, and geopolitical crises.

The US Federal Reserve (Fed) has adopted an aggressive approach to counter high inflation, raising interest rates several times and strengthening the US dollar starting from March 2022 as shown in Fig. 2.

In contrast, the European Central Bank (ECB) was slower to raise rates. By reacting several months later (July 2022), the ECB contributed to relative euro weakness as shown in Fig. 3.

The divergence in monetary policy between the Fed and the ECB

has had significant implications for the financial markets, particularly in the currency exchange sphere. This disparity was not merely a matter of timing; it also reflected differing assessments of economic conditions and inflationary pressures in the respective economies.

The Fed's proactive measures were in response to inflation rates that were rising faster and earlier than in the Eurozone. By raising rates, the Fed aimed to cool down the economy by making borrowing more expensive, thereby reducing spending and investment, and ultimately, slowing inflation. As investors sought the higher yields now available from dollar-denominated assets, the US dollar gained in value.

The ECB's more cautious stance may be attributed to a variety of factors, including a more tempered inflationary environment at the outset, concerns about stifling economic growth, or structural differences in the Eurozone economy. The delay in rate increases, however, meant that returns on Euro-denominated assets were less attractive relative to their US counterparts, which, coupled with investor sentiment, put downward pressure on the Euro.

The implications of these policy choices are far-reaching. A stronger dollar impacts global commodity prices, given that many are priced in dollars, and can put emerging market economies under strain as their dollar-denominated debts become more expensive to service. For the Eurozone, a weaker Euro could potentially boost exports by making them cheaper on the global market, but it also makes imports, including crucial energy supplies, more expensive.

In 2022 the US economy has shown relative resilience in the face of global economic challenges compared to the Eurozone economy. This robustness has attracted investment into dollar-denominated assets, supporting its value. In fact, the energy crisis in Europe, exacerbated by the war in Ukraine, had a significant impact.

European countries are to some extent dependent on Russian resources and specially gas supplies. While countries like Spain, Portugal and England are less vulnerable, benefiting from gas fields in the North Sea and Algeria. Countries like Germany, Italy, Poland, Czech Republic, and many others, are heavily dependent. Their industries are massively impacted, causing production costs to soar and instability to weigh on their economic capacity.

This situation has led to economic uncertainty in Europe, reducing confidence in the euro. Fears of recession in Europe, coupled with high inflation, also played a role in the euro's depreciation against the dollar.

Finally, differences in economic growth prospects between the USA and Europe influenced investor expectations. Moreover, the US has taken advantage of this situation by greatly increasing its exports to Europe, especially of LNG. While the US seemed better positioned to manage global economic challenges, the gloomier outlook for the European economy contributed to the euro's weakness.

Together, these factors led to a significant depreciation of the euro against the US dollar in 2022.

4 Conclusion

The 2022 analysis of the EUR/USD exchange rate highlights a Euro depreciation against the US Dollar, driven by divergent monetary policies, geopolitical events like the energy crisis, and the war in Ukraine.

Investors face challenges with global commodity price shifts and reduced confidence in European markets, impacting trade dynamics. Policymakers must address concerns related to inflation, trade balances, and economic growth. Businesses, especially in Europe, grapple with higher production costs and decreased competitiveness. Acknowledging complexities, the analysis has limitations, including the dynamic nature of geopolitical events. Future research could explore specific effects of crises on exchange rates and assess the long-term impact of divergent monetary policies on global economic stability.

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5 Appendices



Figure 1: Evolution of the EUR/US Exchange Rate in 2022

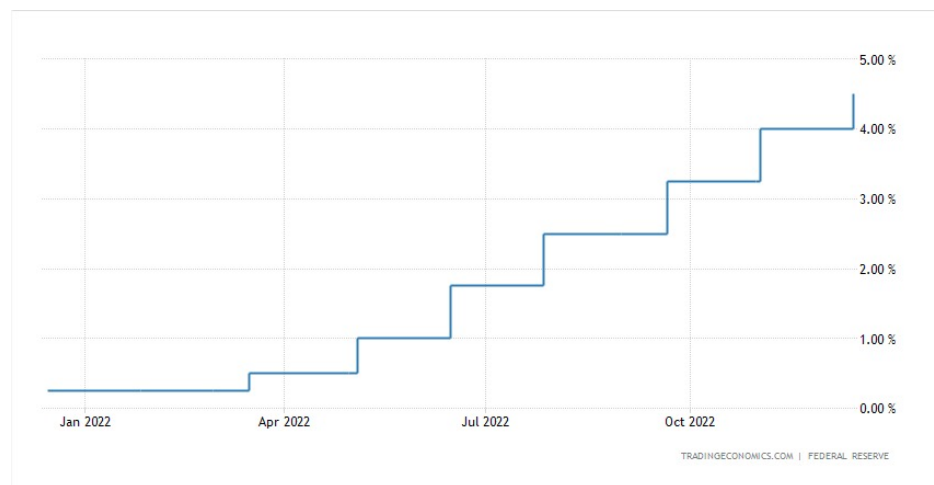


Figure 2: Evolution of the Federal Reserve Interest Rate in 2022

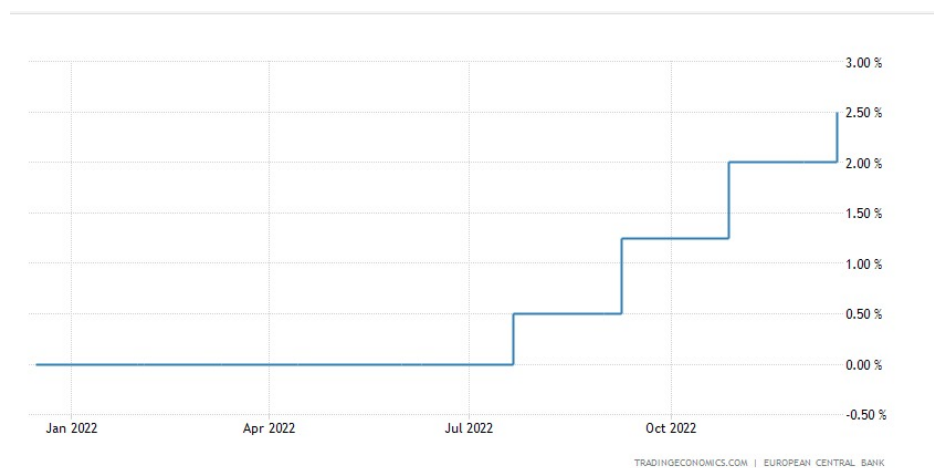


Figure 3: Evolution of the European Central Bank Interest Rate in 2022

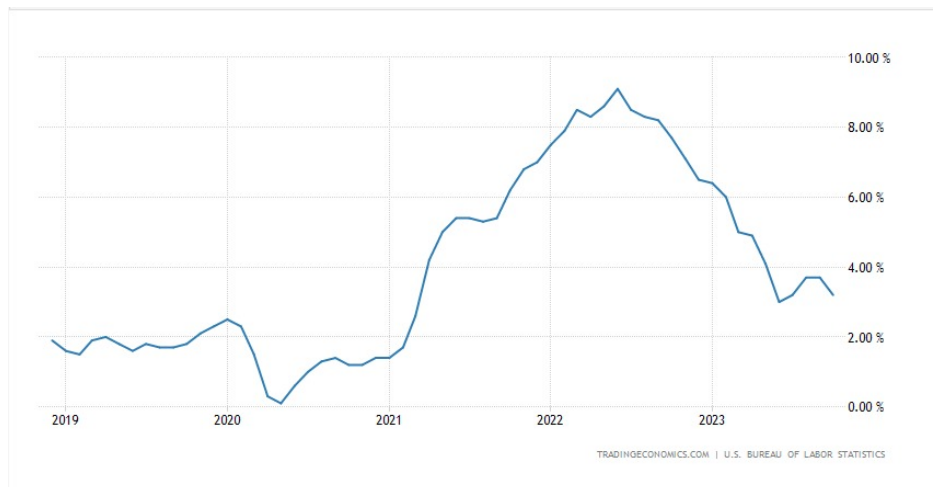


Figure 4: Evolution of the United States Inflation Rate from 2019 to 2022



Figure 5: Evolution of the Euro Area Inflation Rate from 2019 to 2023

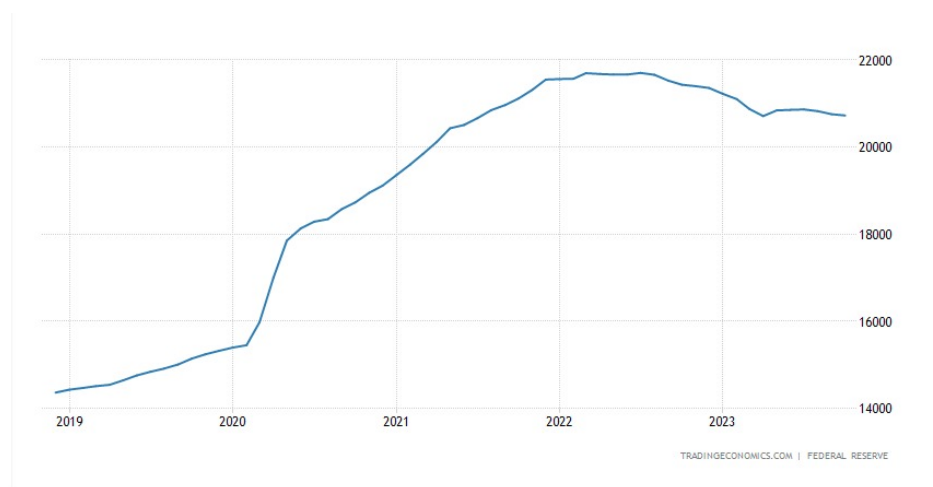


Figure 6: Evolution of the United States M2 Money Supply from 2019 to 2023

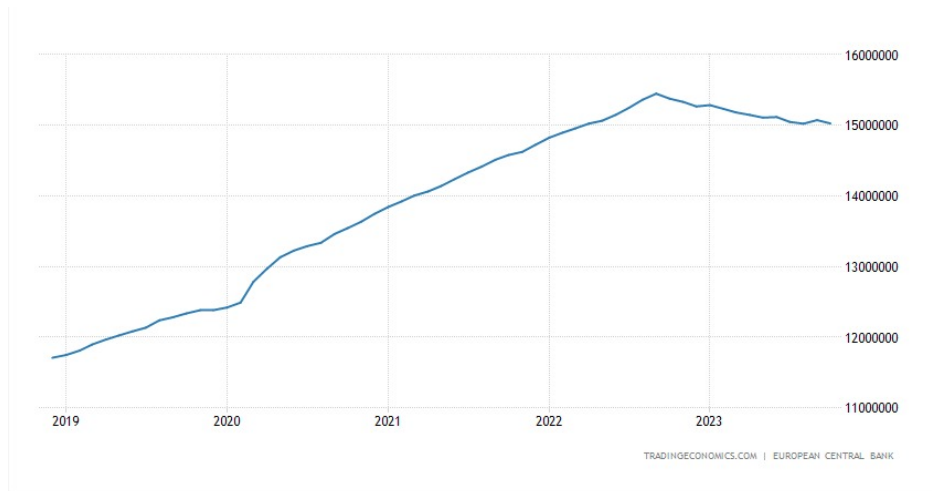


Figure 7: Evolution of the Euro Area M2 Money Supply from 2019 to 2023

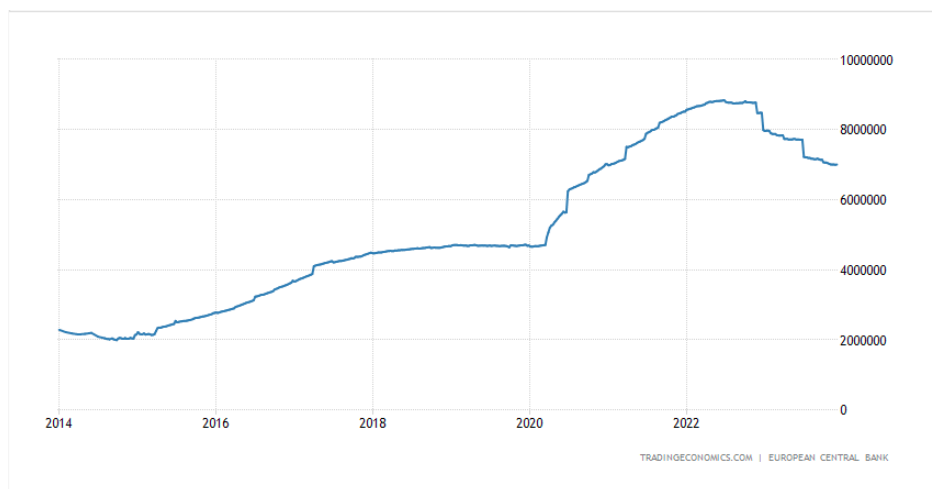


Figure 8: Evolution of the ECB balance sheet

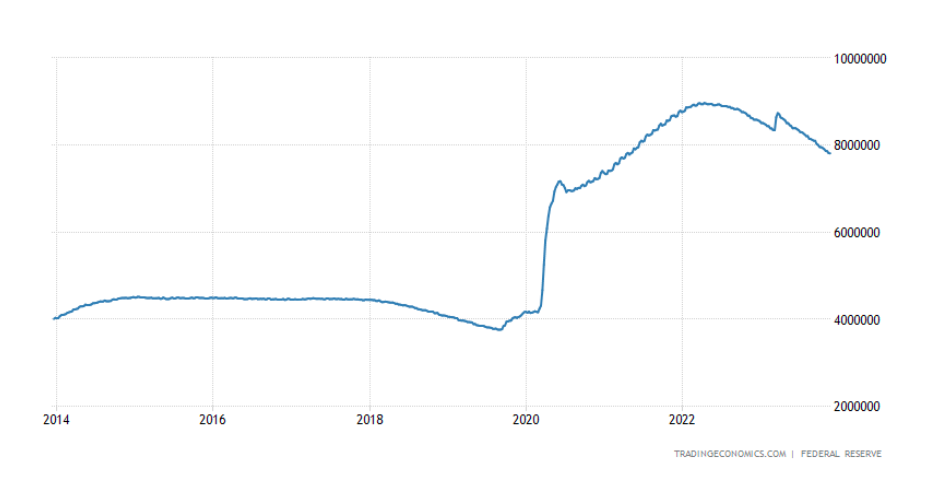


Figure 9: Evolution of the FED balance sheet